When Structure Does Not Help: A Diagnostic Study of Training-Free Gates for Compositional Image–Text Matching

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Abstract

Compositional image–text matching benchmarks ask whether a vision–language model binds attributes, relations, and word order rather than merely recognizing the right objects. Because caption-to-atom QA is already close to TIFA, Davidsonian scene graphs, and recent training-free matching work, we test a narrower reliability hypothesis: dependency-aware atom evidence should act only as a selective veto around holistic VQAScore, with OpenCLIP admitted when the holistic margin is weak. The fixed pairwise protocol scores each image against the benchmark positive and hard-negative captions across ARO, SugarCrepe, and SugarCrepe++ using frozen OpenCLIP, Qwen2.5-VL VQAScore, atom soft-AND, and dependency ablations, with no training, sampling, or threshold tuning on final labels. On 61,902 canonical tasks, the simple VQAScore + CLIP fallback reaches 58,999 correct decisions (0.953), while the reliability gate reaches 58,318 (0.942); paired discordance is one-sided, with 0 gate-only wins and 681 fallback-only wins. Error analysis shows that the gate often blocks small useful CLIP corrections when atom margins are slightly negative, suggesting that generated atom/dependency evidence needs independent calibration before it can safely veto a strong holistic image-conditioned scorer in benchmark-facing matching.

1 Introduction

Compositional image–text matching is deliberately adversarial to global caption similarity. The positive and negative captions often mention the same objects but differ in attribute binding, relation direction, or word order. Winoground, ARO, and SugarCrepe were built to expose these failures rather than to measure generic retrieval fluency (Thrush et al., 2022; Yuksekgonul et al., 2022; Hsieh et al., 2023). The motivating question for this project was whether a lightweight, training-free scorer could use structure to repair those cases without tuning model weights.

The obvious structural story is not new enough to stand alone. TIFA and Davidsonian Scene Graphs already translate captions into question-answerable atoms for fine-grained image evaluation (Hu et al., 2023; Cho et al., 2023). VQAScore offers a strong counterpoint: a single holistic question can be a competitive faithfulness signal and can outperform decomposed question generation in some settings (Lin et al., 2024). ComCLIP and subsequent training-free or inference-time methods further crowd the image–text matching space with causal, entity-grounded, NLI, relation-aware, and local-alignment variants (Jiang et al., 2022; Cascante-Bonilla et al., 2024; Vongala et al., 2025; Zhang et al., 2025; Udai et al., 2025; Miranda et al., 2025). The only defensible thesis was therefore not that decomposition is novel, but that structured evidence might be useful as a selective reliability signal.

We evaluate that thesis directly. The proposed reliability gate keeps holistic VQAScore as the default caption-pair decision. It consults OpenCLIP and dependency-aware atom margins only when the holistic score is low or negative, and it overrides only when the auxiliary signals agree. This design was meant to avoid the known failure mode of soft-AND atom aggregation: noisy generated atoms can dominate an otherwise correct holistic score.

The completed evidence does not support the intended positive claim. Across 61,902 canonical source-backed tasks, the VQAScore + CLIP fallback is stronger than the reliability gate by 681 decisions, with no countervailing gate-only wins. This turns the paper from a method-success claim into a diagnostic negative result: when a strong holistic VQA score is available, the particular dependency/atom gate tested here damages rather than improves the fallback. Winoground remains excluded from measured claims because the official rows were outside the executable benchmark scope under the access constraints of this study.

The paper makes three calibrated contributions. First, it reports a full source-backed negative result for a plausible training-free structured gate rather than polishing a contradicted thesis. Second, it provides paired evidence showing that the fallback dominates the gate on ARO, SugarCrepe, and SugarCrepe++ diagnostics. Third, it analyzes the 681 fallback-only wins to identify where atom and dependency evidence fails to add reliable signal. These contributions are diagnostic; they do not establish a new state of the art.

2 Related Work

Compositionality benchmarks. CLIP-style contrastive pretraining gives a strong global image–text score (Radford et al., 2021), but global scores can behave like bags of words when captions differ only in relation or role assignment. Winoground probes visio-linguistic composition with paired captions and paired images (Thrush et al., 2022). ARO targets attribute, relation, and order sensitivity and explicitly frames bag-of-words behavior as a benchmark failure (Yuksekgonul et al., 2022). SugarCrepe and SugarCrepe++ refine this line by reducing obvious text-only shortcuts and adding lexical/semantic alteration diagnostics (Hsieh et al., 2023; Dumpala et al., 2024). Follow-up audits show that even improved benchmarks need shortcut checks (Udandarao et al., 2025). Other compositional probes stress overlapping but distinct phenomena: VL-CheckList separates object, attribute, and relation checks (Zhao et al., 2022); SVO-style probes target verb and role understanding (Hendricks and Nematzadeh, 2021); CREPE and EqBen expose harder semantic and equivariance failures (Ma et al., 2023; Wang et al., 2023); and hard positive analyses warn that near-miss positives can change conclusions about compositional ability (Kamath et al., 2024). These benchmarks motivate paired evaluation rather than unpaired retrieval summaries, but they also make the evidence standard stricter: a gate must beat strong paired baselines on the same examples, not just produce interpretable intermediate evidence.

Vision–language backbones and scoring signals. The image–text matching literature includes both dual-encoder and fused encoder families. ALIGN, LiT, and CoCa scale contrastive or contrastive-caption training around image/text embeddings (Jia et al., 2021; Zhai et al., 2022; Yu et al., 2022). Fused or generative pretraining systems such as ViLBERT, VisualBERT, LXMERT, UNITER, Oscar, and ALBEF provide context for why a VQA-style score can encode interactions that a global CLIP score misses (Lu et al., 2019; Li et al., 2019; Tan and Bansal, 2019; Chen et al., 2020; Li et al., 2020, 2021). BLIP, BLIP-2, SimVLM, ViLT, and OFA continue this line with unified or generative pretraining objectives that make image-conditioned question scoring practical at inference time (Li et al., 2022, 2023; Wang et al., 2022b; Kim et al., 2021; Wang et al., 2022a). CLIPScore showed that CLIP similarity can be useful as a reference-free caption evaluation signal (Hessel et al., 2021). The present experiment therefore treats CLIP not as a weak strawman, but as a complementary fallback whose margin can correct uncertain holistic VQA decisions.

Training-free matching methods. ComCLIP studies training-free compositional image–text matching with localized components and Winoground-style evaluation (Jiang et al., 2022). Later work adds entity grounding, attribute binding, relation-aware alignment, global-local decoupling, bidirectional retrieval, and inference-time structure (Vongala et al., 2025; Zhang et al., 2025; Udai et al., 2025; Hu et al., 2025; Miranda et al., 2024, 2025; Berasi et al., 2025). Fine-tuning and data-synthesis alternatives such as CLIC and multimodal synthetic compositional data are important context but do not satisfy the training-free constraint of this project (Peleg et al., 2025; Li and Li, 2025). This density of close prior work is why the present paper treats the gate as an empirical hypothesis, not as a broad novelty claim. A negative outcome remains informative because it tests a common design temptation: add structure only at test time, preserve the frozen models, and assume the intermediate atoms are more reliable than the end-to-end score.

VQA, decomposition, and contrastive checks. TIFA and Davidsonian Scene Graphs establish the caption-to-atoms plus VQA pattern for fine-grained visual faithfulness (Hu et al., 2023; Cho et al., 2023). VQAScore instead uses a holistic yes/no prompt and reports strong alignment performance across compositional prompts (Lin et al., 2024). CECE and related NLI-style work use entailment and contradiction generation as another way to

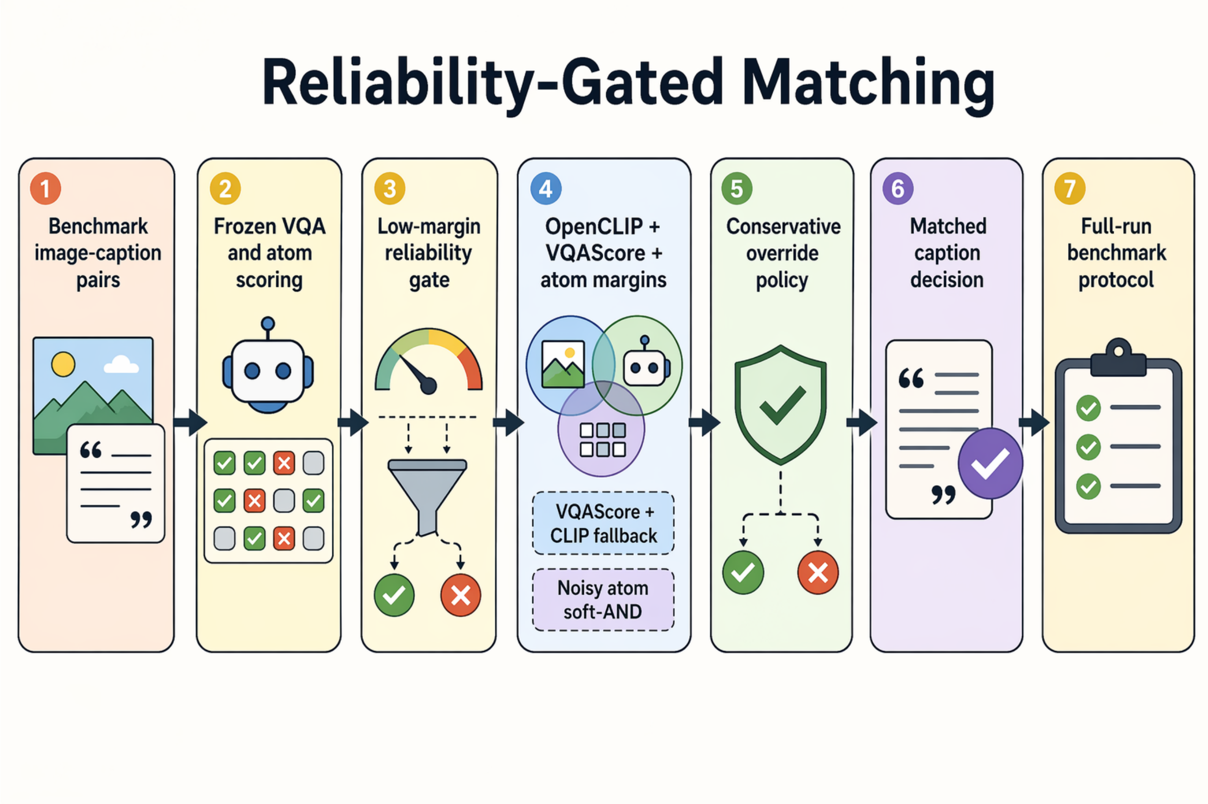

Figure 1: Training-free gate evaluated in this paper. The overview shows the intended scoring policy; the canonical results in Tables 1 and 2 show that the policy is weaker than the VQAScore + CLIP fallback.

stress compositional consistency (Cascante-Bonilla et al., 2024). These papers motivate an evaluation question: if decomposition and contrastive self-checking are already known and noisy, can they still help as a selective gate? Our answer for the implemented gate is no.

3 Task and Method

Each executable benchmark row contains an image, a benchmark-designated positive caption, and a hard negative caption. A method succeeds when it assigns the positive caption a higher image-conditioned score than the negative caption. SugarCrepe and SugarCrepe++ supply this pair format directly; ARO rows are materialized into the same positive-first convention while retaining the official false/true ordering in provenance. Winoground is not included in the paper-facing result table because the official rows were not executable under the documented access constraints for this study.

The evaluated methods are frozen inference-time scorers. OpenCLIP provides the contrastive image–text similarity baseline. Holistic VQAScore asks whether the image shows the complete caption and uses the yes probability as a score. Atom soft-AND decomposes a caption into entity, attribute, and relation checks and aggregates them directly. Dependency-flat and self-negative-off ablations remove role-sensitive structure or generated contrastive checks. The reliability gate combines the holistic score, OpenCLIP margin, and dependency-aware atom margin under a selective override rule.

The atom representation is used as model evidence, not as an oracle. Each caption-level decision retains the original positive and negative captions, the holistic VQAScore margin, the OpenCLIP margin, an atom-derived margin, and the gate reason. The dependency-aware condition preserves subject/object and attribute attachment when the parser/extractor supplies it; the dependency-flat condition intentionally drops that attachment so the comparison isolates whether role-sensitive structure matters. The self-negative-off ablation removes the automatically constructed contrastive checks and therefore tests whether those checks are responsible for any gain or loss.

The rule is intentionally conservative. If holistic VQAScore confidently favors the positive caption, the gate keeps it. If the holistic margin is low or negative, the gate considers auxiliary margins. It changes the decision only when OpenCLIP and atom/dependency evidence both favor the positive caption. The VQAScore + CLIP fallback is simpler: it uses VQAScore as the primary signal and falls back to OpenCLIP in the same low-margin region, without atom/dependency constraints. The central empirical question is whether those extra constraints improve the fallback.

The evaluated-system contract is fixed across rows. Each method receives the same image, positive caption, and hard-negative caption and returns two scalar image-conditioned scores. OpenCLIP-based conditions compute two contrastive image–text similarities with OpenCLIP ViT-B/32. Holistic VQAScore computes two yes-probability scores with Qwen2.5-VL-3B-Instruct under the same declarative caption prompt. The score budget is one frozen score for each caption under each active scorer: two OpenCLIP caption similarities, two VQAScore yes-probability scores, and, for atom conditions, at most eight extracted atom scores per caption. No condition samples candidate captions, tunes thresholds on final labels, or changes the benchmark caption pair.

The fallback and gate reuse those scores under predeclared margin rules. The fallback admits OpenCLIP only when the VQAScore positive-minus-negative margin is at most 0.01 and the OpenCLIP margin is positive. The reliability gate uses the same low-margin region but requires the atom margin to be positive as well; if that atom condition fails, the gate keeps the VQAScore decision. These thresholds are fixed for all ARO, SugarCrepe, and SugarCrepe++ rows, which is why the paired comparison can attribute losses to the structured veto rather than to post-hoc threshold selection.

This comparison is deliberately narrow. We do not claim to reimplement every DSG-, TIFA-, ComCLIP-, or grounding-style method. Instead, the paper tests a specific mechanism that looked plausible after the literature survey: preserve a strong holistic VQA score when it is confident, use CLIP when the holistic margin is weak, and allow generated structure to veto the fallback only when the structure should be diagnostic. Because the full run shows the veto is harmful, the mechanism is discussed as a failed reliability gate rather than as a new compositional matcher.

Why the gate was plausible. The gate was designed around two tensions in the prior work. The first tension is that holistic VQA can score an entire caption in one image-conditioned question, but it can still be uncertain on small relation or attribute changes. The second tension is that atomized questions expose interpretable local checks, but they can introduce parser, question-generation, or VQA noise. A selective gate tries to keep the upside of both: use holistic VQA when it is confident, consult CLIP when the holistic margin is weak, and ask atom/dependency evidence only to block unsafe fallbacks. The completed results show that this last step is exactly where the design fails.

Decision variables. Every final decision is analyzed by method, benchmark family, task identifier, predicted caption index, gold caption index, correctness, scalar score, and gate branch. The overlap analysis bins the same rows by VQAScore margin, OpenCLIP margin, atom margin, gate reason, and text-only behavior. These variables let the paper distinguish a broad aggregate loss from the more specific claim that the loss is concentrated in kept-VQAScore states with useful CLIP corrections.

4 Experimental Setup

Benchmarks and access. The completed matrix covers ARO, SugarCrepe, and SugarCrepe++ replace-relation diagnostics. ARO contributes 52,985 relation, attribution, and order rows. SugarCrepe contributes 7,511 official refined hard-negative rows. SugarCrepe++ contributes 1,406 replace-relation diagnostic rows. These are real public benchmark sources with downloaded or verified image assets and fixed positive/negative semantics. Winoground remains an intended benchmark but is not paper-facing here because its official examples were not executable under the access constraints for this study.

The three executable families are not treated as interchangeable slices. ARO is large and relation-heavy, so it dominates aggregate counts and exposes whether a method behaves like a bag-of-words scorer at scale. SugarCrepe is smaller but was designed to reduce obvious caption-only hacks, making it a useful check against inflated gains from textual shortcuts. SugarCrepe++ replace-relation rows stress whether lexical or semantic alterations destabilize the scorer. This is why the main table reports both family-level and aggregate numbers, and why the paired table repeats the gate-vs-fallback comparison within each family.

Models and metrics. The paper-facing backends are OpenCLIP ViT-B/32-style contrastive scoring and VQAScore with Qwen2.5-VL-3B-Instruct. All methods are evaluated on the same 61,902 canonical semantic tasks after duplicate raw rows are audited and deduplicated by method-task key. The metric is pairwise accuracy: the positive caption must receive the higher score. For the gate-vs-fallback comparison, the same task set supports paired discordance counts and McNemar-style tests.

The evaluated method set is intentionally broader than the final diagnostic claim. OpenCLIP and holistic VQAScore are the primary baselines. VQAScore + CLIP fallback is the strongest non-structured control and therefore the required comparison for the reliability gate. Atom soft-AND, dependency-flat, self-negative-off, relation-residual dependency, and text-only diagnostic rows serve as ablations or sanity checks. The text-only row is not a proposed method; it is included to detect whether a pattern could be explained without image evidence.

Integrity policy. Metric reporting uses a canonical deduplicated row basis: 557,118 method-task decisions across nine methods, with 61,902 semantic tasks per method. A duplicate check found 24 repeated observations and no scoring-contract conflict, so each reported metric uses one decision per method-task pair. This matters for the negative result: the fallback win is not a denominator effect, and every comparison in Table 2 is paired on the same semantic tasks.

The run contract also fixes how missing evidence is discussed. Winoground is not backfilled with a proxy group score, because the official examples were not available under the benchmark access constraints. The cached-score abstention idea is not promoted to a result, because no held-out calibration threshold was declared before inspecting final labels. These decisions reduce the apparent scope of the paper, but they keep the scientific claim tied to executed, reproducible rows.

Canonicalization and paired testing. The canonicalization step removes repeated observations while preserving the underlying semantic example and method score. The duplicate audit found 24 extra rows and no scoring-contract conflicts, so metric tables use one canonical row per method-task pair. The paired gate-vs-fallback test then joins the two methods by the same task identifier and counts only discordant outcomes. The reported 681-to-0 discordance is therefore a same-example comparison, not a comparison between differently filtered subsets.

Stopping and exclusion rules. All method conditions share the same 61,902 executable semantic tasks and are included only after every required method has a canonical row. Rows from benchmark construction, smoke runs, pilot runs, or inaccessible official splits are not promoted into the paper tables. This rule is conservative for the present paper because it excludes the originally targeted Winoground group score, but it prevents a stronger-looking story from being assembled out of non-comparable evidence.

5 Results

The main result is unambiguous. VQAScore + CLIP fallback is the strongest evaluated condition on all three executable benchmark families and on the 61,902-task aggregate. It reaches 58,999 correct decisions overall (0.953). The reliability gate reaches 58,318 correct decisions (0.942), below the fallback by 681 decisions. This means the gate does not merely fail to improve the fallback; it loses every paired disagreement observed in the canonical comparison.

The remaining baselines clarify the scale of the failure. Holistic VQAScore is already much stronger than OpenCLIP alone, so the fallback's value comes from a limited set of low-margin corrections rather than from replacing VQA globally. Atom soft-AND stays close to OpenCLIP, which means direct atom aggregation is not an adequate substitute for the holistic question. The dependency-residual variant is worse still, so relation-aware residuals in this implementation do not provide a hidden rescue path. These comparisons are why the paper treats the fallback as the relevant control and the gate as an unsuccessful attempt to regularize that control.

Table 2 shows the paired shape of the failure. On the full matrix, gate-only wins are 0 and fallback-only wins are 681. The same one-sided pattern appears within ARO, SugarCrepe, and SugarCrepe++ separately. The positive reliability-gate thesis is therefore removed from the paper rather than softened into a weaker success claim.

The paired design is important because the aggregate family sizes are uneven. ARO contributes most tasks, but the gate does not recover a favorable

Method and role	Scorer / policy	ARO (52,985)	SugarCrepe (7,511)	SugarCrepe++ (1,406)	Overall (61,902)
OpenCLIP pair scorer (baseline)	ViT-B/32 caption similarities	31,137 (0.588)	5,901 (0.786)	1,015 (0.722)	38,053 (0.615)
Holistic VQAScore (baseline)	Qwen2.5-VL yes-probability scores	47,686 (0.900)	7,159 (0.953)	1,267 (0.901)	56,112 (0.906)
VQAScore + CLIP fallback (control)	Use CLIP if VQA margin ≤ 0.01 and CLIP >0	**50,317 (0.950)**	**7,342 (0.978)**	**1,340 (0.953)**	**58,999 (0.953)**
Reliability gate (tested)	Fallback plus atom margin >0 veto	49,692 (0.938)	7,310 (0.973)	1,316 (0.936)	58,318 (0.942)
Atom soft-AND (ablation)	OpenCLIP atom scores, soft conjunction	30,849 (0.582)	5,990 (0.797)	947 (0.674)	37,786 (0.610)
Dependency-flat (ablation)	Atom scores without dependency roles	30,887 (0.583)	5,968 (0.795)	924 (0.657)	37,779 (0.610)
Self-negative off (ablation)	Dependency atoms without self-negatives	31,206 (0.589)	5,949 (0.792)	1,018 (0.724)	38,173 (0.617)
Relation residual (diagnostic)	Dependency residual with self-negative term	10,731 (0.203)	3,506 (0.467)	475 (0.338)	14,712 (0.238)
Text-only diagnostic	Caption-only heuristic; no image score	41 (0.001)	2,871 (0.382)	662 (0.471)	3,574 (0.058)

Table 1: Main result over three source families: under the fixed pairwise-accuracy protocol described in Section 4, VQAScore + CLIP fallback is strongest on 61,902 canonical tasks (58,999 correct; 0.953), while the reliability gate is lower (58,318 correct; 0.942).

Family	System	N	Correct	Only wins
All	Gate	61,902	58,318	0
All	Fallback	61,902	**58,999**	**681**
ARO	Gate	52,985	49,692	0
ARO	Fallback	52,985	**50,317**	**625**
SugarCrepe	Gate	7,511	7,310	0
SugarCrepe	Fallback	7,511	**7,342**	**32**
SugarCrepe++	Gate	1,406	1,316	0
SugarCrepe++	Fallback	1,406	**1,340**	**24**

Table 2: Paired comparison: fallback-only wins are 681 versus 0 gate-only wins over the full matrix, contradicting a gate-improvement claim.

slice on the smaller SugarCrepe or SugarCrepe++ families. If the reliability mechanism were merely over-specialized to ARO relation rows, one would expect at least some gate-only wins on the refined hard-negative benchmarks. The recorded discordance shows the opposite: every observed disagreement favors the simpler fallback.

This result also rules out two weaker positive interpretations. One would be to claim that atom/dependency evidence is useful whenever holistic VQAScore fails. The paired table rejects that interpretation because the observed corrections come from the fallback condition, not from the structured gate. Another would be to claim that the gate is a harmless interpretability layer. The main table rejects that framing as well: adding the gate changes decisions and reduces accuracy. Interpretability is not free when intermediate evidence is allowed to veto a stronger scorer.

6 Analysis

The override-state breakdown explains where the design fails. Across 61,902 task-aligned rows, the reliability policy entered the explicit override state for 2,637 rows and kept VQAScore for 59,265 rows. The override rows do not produce fallback-only losses; their fallback-only count is 0. The damage comes from the kept-VQAScore state, which accounts for 681 fallback-only wins. In other words, the tested atom/dependency constraints do not rescue the fallback in the cases where it matters; they instead prevent the simpler fallback from using OpenCLIP in low-margin holistic cases.

This distinction changes the interpretation of the gate. A useful reliability gate should identify cases where the primary scorer is unreliable and then admit a better auxiliary signal. The implemented gate identifies many low-margin cases, but its atom/dependency veto is more conservative than useful. Because the override rows do not explain the losses, merely relaxing the override threshold is not a principled fix. The failure is located in the decision to trust slightly negative atom margins as evidence against the CLIP correction.

The overlap table accounts for all 681 fallback-only wins. The largest groups come from Visual Genome relation and attribution rows inside ARO. In the largest cluster, 320 relation examples keep the VQAScore decision even though the holistic VQA margin is below -0.05, OpenCLIP gives a small positive margin, the atom margin is slightly negative, and the text-only diagnostic is wrong. The next largest clusters repeat the same pattern: 101 attribution examples with a VQA margin below -0.05, 95 relation examples with a VQA margin between -0.05 and 0, and 60 attribution examples with a VQA margin between -0.05 and 0. These groups share a common pattern: VQAScore margins are negative or near zero, OpenCLIP margins are small but positive, atom margins are slightly negative, and text-only behavior usually fails. This is exactly the regime where a selective gate should help if the atoms are reliable. Instead, the atom margin acts as a veto against the simpler fallback's useful correction.

The text-only diagnostic is also informative. The largest fallback-only groups are not solved by text alone, which weakens the concern that the fallback is winning only because the negative captions contain obvious shortcuts. Instead, the pattern is image-conditioned but fragile: OpenCLIP supplies a small positive margin, VQAScore is uncertain or wrong, and the generated atom margin is just negative enough to block the correction. That is a narrower and more useful failure story than a generic statement that decomposition is noisy.

The ablations support the same diagnosis. Atom soft-AND trails holistic VQAScore and the fallback, so direct decomposition is not competitive as a main scorer. The current relation-residual dependency scorer collapses relative to every strong image-conditioned baseline. Self-negative-off is close to OpenCLIP rather than to the VQAScore fallback, which means generated contrastive checks do not explain the strong fallback result. A cached-score abstention-only gate is left as exploratory because no threshold rule was predeclared on a held-out calibration split.

A predeclared pivot would therefore need a different evidence contract. One option is to learn or tune a calibration policy on a held-out split and test it once on the final matrix. Another is to validate atom quality independently, for example by filtering atoms whose question form or dependency attachment is unstable before they can veto a fallback. A third is to compare against full external implementations of ComCLIP, DSG-style QA, TIFA-style QA, or grounding methods. None of those repairs is claimed here because the current draft is restricted to the completed canonical run.

What the negative result does and does not say. The result does not say that all structured inference-time methods are unhelpful. It says that this specific reliability gate, evaluated without training or post-hoc threshold tuning, is not a safe improvement over a VQAScore-first fallback. That distinction matters for future work. A structured method could still help if it uses better localization, a learned calibration layer, a held-out threshold policy, or a different decomposition quality gate. The present evidence only licenses a narrower operational lesson: do not let generated atoms veto a useful CLIP correction unless the atom scorer has been validated on the same type of hard-negative pair.

Why a negative paper is still useful. The failed gate is not a random poor baseline. It is a direct instantiation of a common intuition: if a caption pair differs in a relation or attribute, parse the caption into atomic checks and require those checks to agree before trusting a fallback. The literature review explains why this intuition is attractive, and the full run explains why it is risky. The value of the paper is therefore diagnostic: it records an evidence boundary that can prevent future work from presenting uncalibrated atom evidence as reliability improvements.

7 Failure Cases

The negative result has three practical failure modes. First, atom questions are not gold semantic evidence. A relation or attribute atom can be underspecified, attached to the wrong participant, duplicated, or scored inconsistently by the frozen VQA backend. Second, dependency constraints can be too brittle for a pairwise decision: a small negative atom margin can block a useful OpenCLIP fallback even when the holistic score is uncertain. Third, benchmark shortcuts remain a concern, but the text-only diagnostic is not strong enough here to explain the fallback's success. The failure is therefore not that a caption-only heuristic solved the task; it is that the tested structured veto was less reliable than the simpler VQAScore/OpenCLIP rule.

The first failure mode is parser/extractor uncertainty. When an atom turns a caption relation into a yes/no question, the question may omit the very role that distinguishes the positive and negative captions. A dependency edge can make the representation look precise while still preserving the wrong attachment. In pairwise evaluation, that error is costly because the method needs only a relative margin, not an absolute explanation.

The second failure mode is calibration mismatch across scorers. VQAScore, OpenCLIP, and atom scores do not share a common probability scale. A slightly negative atom margin can be less reliable than a slightly positive CLIP margin, but the gate treats the atom margin as a veto. The full matrix shows this calibration choice is harmful under the implemented thresholds.

The third failure mode is benchmark-level heterogeneity. ARO, SugarCrepe, and SugarCrepe++ stress different hard-negative constructions. A rule that appears safe on relation-heavy rows can still fail on refined hard negatives and lexical/semantic alteration rows. Reporting all families separately prevents the large ARO denominator from hiding the fact that the gate has no positive paired slice in the executable benchmark set.

The largest fallback-only clusters make these failure modes concrete. Relation rows account for 320 of the failures in the most common analysis bin and another 95 in a nearby low-margin bin. In both bins, the holistic VQA score is negative or nearly tied, OpenCLIP gives the positive caption a small advantage, and the atom margin is just below zero. The gate interprets that atom margin as a reason to keep or protect the VQAScore decision. The fallback instead treats the same state as an admission case for CLIP and is correct.

Attribution rows show the same calibration problem rather than a separate relation-only weakness. The two largest attribution bins contribute 101 and 60 fallback-only wins under the same score pattern: uncertain VQAScore, weakly positive OpenCLIP, slightly negative atom evidence, and a failed text-only diagnostic. That combination matters because it rules out the easiest explanation for the fallback's advantage. The fallback is not simply exploiting caption shortcuts; it is using an image-conditioned margin that the structured veto blocks.

These row-level facts explain why the paper does not propose a threshold tweak as a hidden positive result. A lower veto threshold could recover some of the 681 fallback-only wins, but it would be chosen after observing final labels and would no longer test the original predeclared reliability gate. The stronger conclusion is narrower: generated atom evidence needs an independent calibration contract before it can safely veto a simpler image-conditioned fallback.

Implications for future gates. Future gates should separate three responsibilities that the current policy conflates. Detection asks whether the holistic scorer is uncertain. Admission asks whether a fallback has enough independent image-conditioned evidence to be used. Veto asks whether structured evidence is reliable enough to block that fallback. The present results suggest that detection and admission can be implemented with VQAScore and CLIP margins, but veto requires a stronger validation contract. Without that contract, structure becomes an extra failure surface rather than a reliability layer.

8 Conclusion

This paper reports a negative diagnostic result for a plausible training-free compositional matching gate. The implemented dependency-aware atom gate does not improve over a VQAScore + CLIP fallback on the completed public benchmark matrix. It records 0 gate-only wins and 681 fallback-only wins over 61,902 canonical tasks, so the original positive method claim is contradicted. The useful outcome is the boundary condition: when holistic VQA and CLIP already provide complementary margins, generated atom/dependency evidence must be validated as a calibration signal before it is allowed to veto the fallback.

Limitations

The study evaluates one implemented gate and should not be read as a theorem against all structured inference-time methods. A held-out calibration split, stronger atom validation, or external comparators such as faithful ComCLIP, DSG-style, TIFA-style, or recent grounding baselines could support a new predeclared pivot. Winoground official group-score evidence remains absent because the official rows were outside the executable benchmark scope of this study. The paper also does not claim that the benchmark families cover all forms of multimodal compositionality.

Ethical Considerations

This work uses public or access-controlled research benchmarks under their documented access constraints. It makes no deployment or safety-certification claim. Reporting the negative result is important because inference-time structure can look interpretable while still reducing accuracy. Generated atom outputs should therefore be treated as model evidence, not as human-verified explanations.

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A Reproducibility Notes

The canonical run is the full source-backed matrix over ARO, SugarCrepe, and SugarCrepe++ diagnostics. The paper-facing regeneration interface is: rebuild the benchmark task files from the documented public sources, execute the nine scoring conditions on the same positive/negative rows, canonicalize duplicate method-task rows with the recorded deduplication contract, regenerate the analysis tables, and rebuild this manuscript. The supplementary materials record provenance, row-count, model/backend, metric, and result details needed to audit the reported benchmark and method semantics.

A reproduction should check five invariants before comparing claims. First, the three executable source families must contain 52,985 ARO rows, 7,511 SugarCrepe rows, and 1,406 SugarCrepe++ rows after access filtering. Second, every promoted method must emit one canonical decision for each of the 61,902 semantic tasks; smoke rows, construction rows, and inaccessible rows are excluded from paper tables. Third, the fallback and gate thresholds must be the same thresholds used in the main text: VQAScore margin at most 0.01, positive OpenCLIP margin for fallback admission, and positive atom margin for the structured gate. Fourth, duplicate observations must be resolved by the method-task contract before any family or aggregate accuracy is computed. Fifth, the paired gate-vs-fallback comparison must join the two methods on the same task identifiers and report gate-only and fallback-only wins before interpreting the aggregate difference.